

Geometry

Unit 11

Lesson 3 Homework

Name_____

Date_____

Period_____

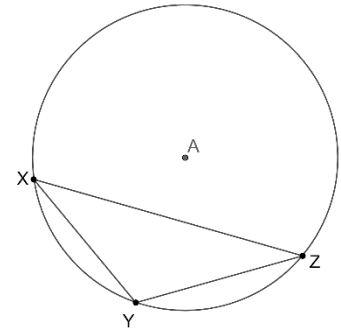
Circle A is shown, where points X, Y , and Z are on circle A and create $\triangle XYZ$. $m\widehat{XZ} = 240^\circ$, $m\angle ZXY = 20^\circ$, and $m\angle XYZ = (4x - 28)^\circ$.

1. Find the value of x .

2. Find the $m\angle XYZ$.

3. Find the $m\widehat{YZ}$.

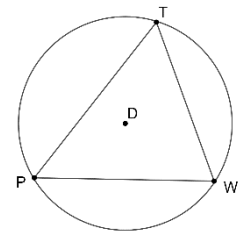
4. Find $m\angle YZX$.



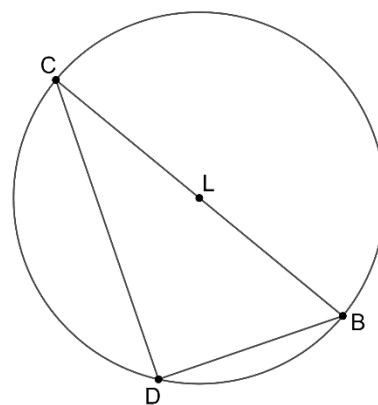
Circle D is shown, where points P, T , and W are on circle D and create $\triangle PTW$. $m\angle PTW = 54^\circ$ and $m\angle TWP = 60^\circ$.

5. Find $m\widehat{PT}$.

6. Find $m\widehat{TW}$.



Circle L has points B, C , and D on circle L and diameter $\overline{BC}$. $\triangle BCD$ is inscribed in the circle. $m\widehat{CD} = 100^\circ$, $m\angle DCB = 40^\circ$, and $m\angle DBC = (2x + 12)^\circ$.



7. What is the $m\angle CDB$?

8. What is the value of x ?

9. Determine $m\angle DBC$.

10. Determine $m\widehat{DBC}$?